Roy **Rajjyoti**

Generative AI - Data Scientist

Bengaluru, Karnataka, India

(+91) 8240781936 | rajjyotiroy15@gmail.com | linkedin.com/in/royrajjyoti

SUMMARY

With 7.5 years of experience in the IT sector I have worked in the domains of Identity and Access Management (IDAM) and Artificial Intelligence (AI), gaining valuable expertise in these fields. For the past 6 years, I have been fully immersed in the exciting world of Data Science, serving as an AI Engineer/Data Scientist. This role has allowed me to leverage my skills in Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Generative AI and Python programming to extract meaningful insights from complex data sets. Currently, I am actively contributing to the Generative AI Research and Product Development Team at Eli Lilly & Company, where I contribute to cutting-edge projects and innovative solutions.

EDUCATION

B, .Tech in Electrical Engineering

Maulana Abul Kalam Azad University of Technology (WBUT) Year of Passing -2018
Kolkata, West Bengal

M.Tech in Data Science and Engineering

Birla Institute Of Technology And Science – Pilani (BITS–Pilani WILP)
Year of Passing- 2024

EXPERIENCE

Generative AI – Data Scientist at Eli Lilly & Company

Experience – December 2023 –Present
Bengaluru, Karnataka

AI Engineer at Jio Platforms Limited (JPL)

Experience – October 2022 –December 2023
Bengaluru, Karnataka

Project Engineer at Wipro Limited

Experience: October 2018- September 2022
Kolkata, West Bengal

SKILLS

Programming Languages

- Python

Machine Learning

Data Wrangling | Data Cleaning| Feature Engineering| ExploratoryData Analysis (EDA)| Classification | Linear Regression | Logistic Regression | Gradient Descent | Clustering | Ensembles| Xgboost| RandomForest | SVM| k-NN | K- means| Data Mining

Tools & Frameworks

Pandas| Numpy | Seaborn| Sklearn | Matplotlib | Microsoft Excel | NLTK | AWS |MLOPS | Amazon Web Services (AWS) | Deepeval | AWS Q | Synthetic Data Generation |

Analytics

Generative AI | Large Language Models (LLMs) | Pytorch | Retrieval Augmented Generation (RAG) | Langchain | LanGraph | OpenAI | Prompt Engineering | Vector database | Agentic AI | Bedrock Models

Statistical & Natural Language Processing

HypothesisTesting | Principal Component Analysis| RegressionAnalysis| T-SNE | Bag of Words| TF-IDF| Word2Vec |

PROJECTS

Lead Data Scientist at Eli Lilly and Company, Bangalore, India *Dec 2023 - PRESENT*
Bengaluru, Karnataka

- Led the development of a dynamic mapping solution for contractor invoices across multiple manufacturing sites by building a mapper that converts diverse invoice fields into standardized canonical fields, enabling scalable processing and improved financial visibility.
- Designed an LLM-powered workflow optimized to minimize LLM calls and processing time, reducing operational costs while integrating with an OCR, classification, and extraction pipeline to detect cost leakage.
- Developed an LLM-based classifier integrated with a RAG pipeline to determine when user queries require LLM-generated responses versus retrieval-based responses.
- Generated and labeled ~3000 synthetic data samples to train and improve the LLM classifier and established a feedback loop for continuous model improvement.
- Built a Citation Checker system that validates model-generated citations against original source documents and generates accuracy scores to detect hallucinated references.
- Developed IR GPT, an AI-powered system that converts large investor relations document repositories into actionable insights through intelligent question answering and hierarchical document summarization.
- Created reusable Python components and internal libraries to standardize frequently used AI/ML workflows and accelerate development across teams.
- Implemented an AWS Q–based enterprise chatbot to replace a legacy chatbot system and improve internal knowledge retrieval and query resolution.
- Mentored junior data scientists and interns while leading exploratory research and Proof-of-Concept (PoC) development for new AI initiatives.

AI Engineer at Jio Platforms Limited (JPL) *Sep 2022- Dec 2023*
Bengaluru, Karnataka

- Developed and implemented a Cloud Workload Prediction system, utilizing advanced data science techniques and machine learning algorithms.
- Generated forecasts for critical resource utilization, including CPU usage, memory consumption, disk usage, and network bandwidth.
- Collaborated closely with the cost management team to optimize resource allocation, resulting in improved cost-effectiveness.
  Actively engaged in Research and Development (R&D) efforts for creating a Proof of Concept (POC) Virtual Try-On technology, enabling customers to virtually try on clothing, makeup, and various products.
- Leveraged cutting-edge machine learning and AI techniques to replicate the in-store fitting experience, enhancing the online shopping experience for users.
- Led the fine-tuning and quantization initiative for Large Language Models (LLMs) to create a POC for an LLM-based chatbot.
- Employed advanced techniques such as Qlora (PEFT Technique) to optimize model performance, reducing computational overhead while maintaining accuracy.
- Successfully optimized LLMs, enhancing their efficiency and responsiveness, ultimately leading to improved system performance and user experience.

Project Engineer at Wipro Limited Oct 2018 - Sep 2022
Kolkata, West Bengal

E-Commerce Product Recommendation System:

- Conducted data cleaning and preprocessing for enhanced data quality.
- Employed text vectorization techniques like Bag of Words (BoW) and tf-idf to create vectors from product descriptions.
- Utilized content-based filtering to improve product recommendations.
- Successfully recommended products with high similarity values, improving user experience and driving sales.

Malware Classification Model:

- Conducted thorough Exploratory Data Analysis (EDA) for insights into malware datasets.
- Extracted essential features from byte files and asm files to enhance model accuracy.
- Implemented three classification models: Logistic Regression, Random Forest Classifier, and XGBoost Classifier.
- Evaluated model performance using Multi-Class Log Loss and Confusion Matrix.
- Conducted a comparative analysis of model results to identify strengths and weaknesses.
- Key Metrics: Multi-class Log Loss, Confusion Matrix.

Privileged Access Management (PAM):

- Led user on-boarding and account on-boarding processes.
- Managed privileged sessions through Privileged Session Manager, ensuring monitoring, control, and isolation.
- Created and enforced password policies in line with Client Password policy.
- Secured and maintained privileged accounts using EPV (CyberArk Enterprise Password Vault)